Analytics Vidhya

Policies and Value Functions

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Policy

A policy π is a distribution over actions, given states, defined as:

$$\pi(a \mid s) = \mathbb{P}[A_t = a \mid S_t = s]$$

- Policy π models the behavior of an agent
- In MDP, policy depends only on the current state
- Policies are stationary
- Given a policy, one can define state-value function and action-value function

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Policy

State-Value Function

In an MDP, the expected return staring from a state s under a policy π is known as the state-value function

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$$

Action-Value Function

In an MDP, the expected return staring from a state s, an action a, and then following the policy π is known as the action-value function $q_\pi(s, a)$ is given by:

$$q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

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Thank You